

Cybersecurity Internship Report

Week 3 – Advanced Security and Final Reporting

Name: Abdul Ahad

Internship: Cybersecurity Analyst Internship

Week: 3 – Advanced Security and Final Reporting

Objective

The objective of Week 3 was to perform basic penetration testing, implement application logging, create a security best practices checklist, and prepare the project for final submission.

1. Basic Penetration Testing

Basic penetration testing was performed to verify that the security measures implemented during Week 2 were functioning correctly.

Tests Performed

✓ Input Validation

- Tested invalid email formats and empty password fields.
- The application successfully rejected invalid input.

✓ JWT Authentication

- Logged in using valid credentials.
- Verified through the browser Developer Tools (**Network** → **Response**) that a JWT token was successfully generated and returned.

✓ Password Hashing

- Created new user accounts ahad@gmail.com with password pass1234 and Cyber@security.com with password password1234, and verified through phpMyAdmin that passwords were stored as **bcrypt hashes** instead of plain text.

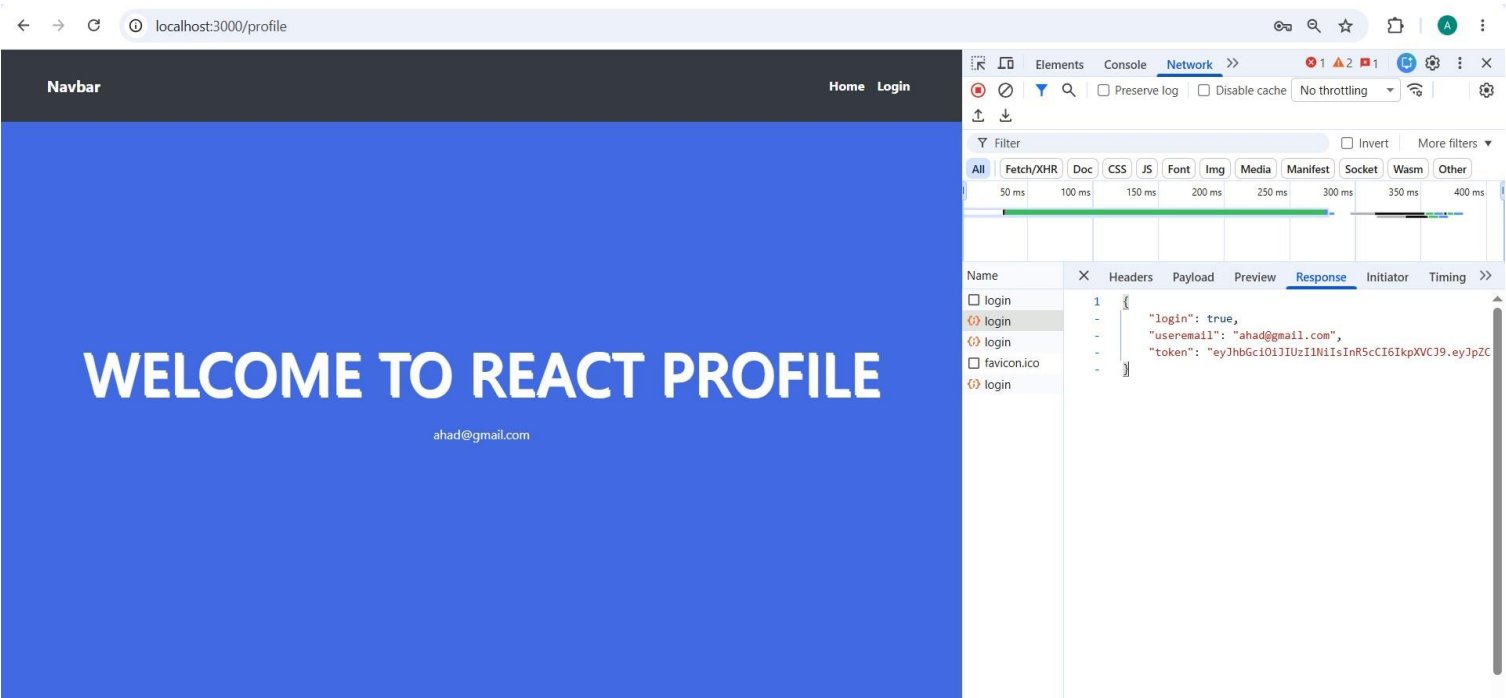
✓ Helmet Security Headers

- Inspected the application's response headers using the browser Developer Tools.
- Confirmed that Helmet successfully added security headers such as **X-Frame-Options** and **X-Content-Type-Options**.

Result

All implemented security mechanisms functioned successfully and improved the application's resistance against common web attacks.

📸 Screenshot 1
JWT Token in Login Response



📸 Screenshot 2
Hashed Passwords in phpMyAdmin

		id	email	password	date
☐	Edit Copy Delete	1	somu@gmail.com	\$2b\$10\$Y6mo4clHl8tHxZ12F3xf.i8CmqNsFUEkxXdCBTzzQW...	2021-06-26 11:37:39
☐	Edit Copy Delete	2	somuu@gmail.com	\$2b\$10\$GVcA24UNAJIV3dU5zgFCRuU4s9bn0nUhaYVZRu.JVzi...	2021-06-26 11:38:13
☐	Edit Copy Delete	3	ram@gmail.com	\$2b\$10\$0bp6mwJ2aJnTh3LuGpJtTuMSVqw1WNQJbK7EL6A8R59...	2021-06-26 11:46:15
☐	Edit Copy Delete	4	atanuj383@gmail.com	\$2b\$10\$NN2z7zs6loysol8OEda1xelBhuM6Crrh/woHyYHzLJe...	2021-06-26 12:22:18
☐	Edit Copy Delete	5	atanuj5@gmail.com	\$2b\$10\$fKTJbChGRpZLgRkXS6moA.zQATJ8jVoBU66wV57vkeE...	2021-06-26 12:27:05
☐	Edit Copy Delete	6	atanuj55@gmail.com	\$2b\$10\$nnbEitRzcfPwd28a7XGc8e00Zvm38sob6i8Bv8nyxOV...	2021-06-26 12:27:55
☐	Edit Copy Delete	7	rajuahamed.kh@gmail.com	\$2b\$10\$pDrNNqQSNibwoH/1fPW9t.9LUgbY28kge4FLcH0BYKh...	2021-06-26 14:45:36
☐	Edit Copy Delete	8	kkk@gmail.com	\$2b\$10\$WvfdXiguQbF8zoSfXy7qGeneV0XGSdglMGriRppaJgi...	2021-06-26 15:02:13
☐	Edit Copy Delete	9	samrat1@gmail.com	\$2b\$10\$dXEjGzCNaPIRybHMRnK06.cbwnA6vTkSjld2q9xKLbD...	2021-06-26 17:45:07
☐	Edit Copy Delete	12	ahad@gmail.com	\$2b\$10\$AotCDflnFeRrbUaT2Ri0.O0jkNi.7xbYlezoQ.LAYOg...	2026-06-27 08:19:45
☐	Edit Copy Delete	13	Cyber@security.com	\$2b\$10\$ZtnWtMYw7I4YoculsyjfK.6NPie09e.U517dLYJ3Hgu...	2026-06-27 13:22:18

☐ Check all

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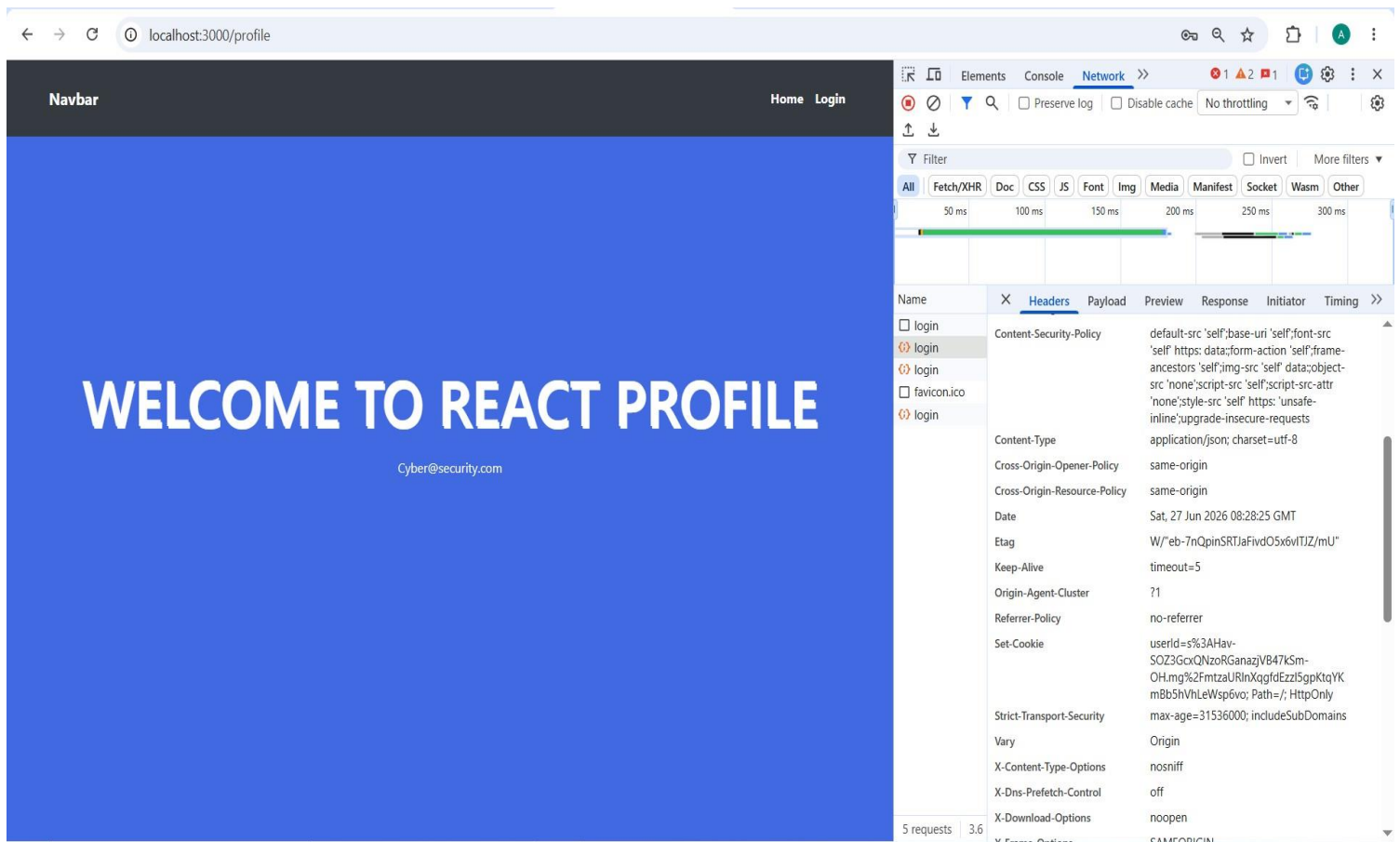
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📸 Screenshot 3

Helmet Security Headers



Basic Logging

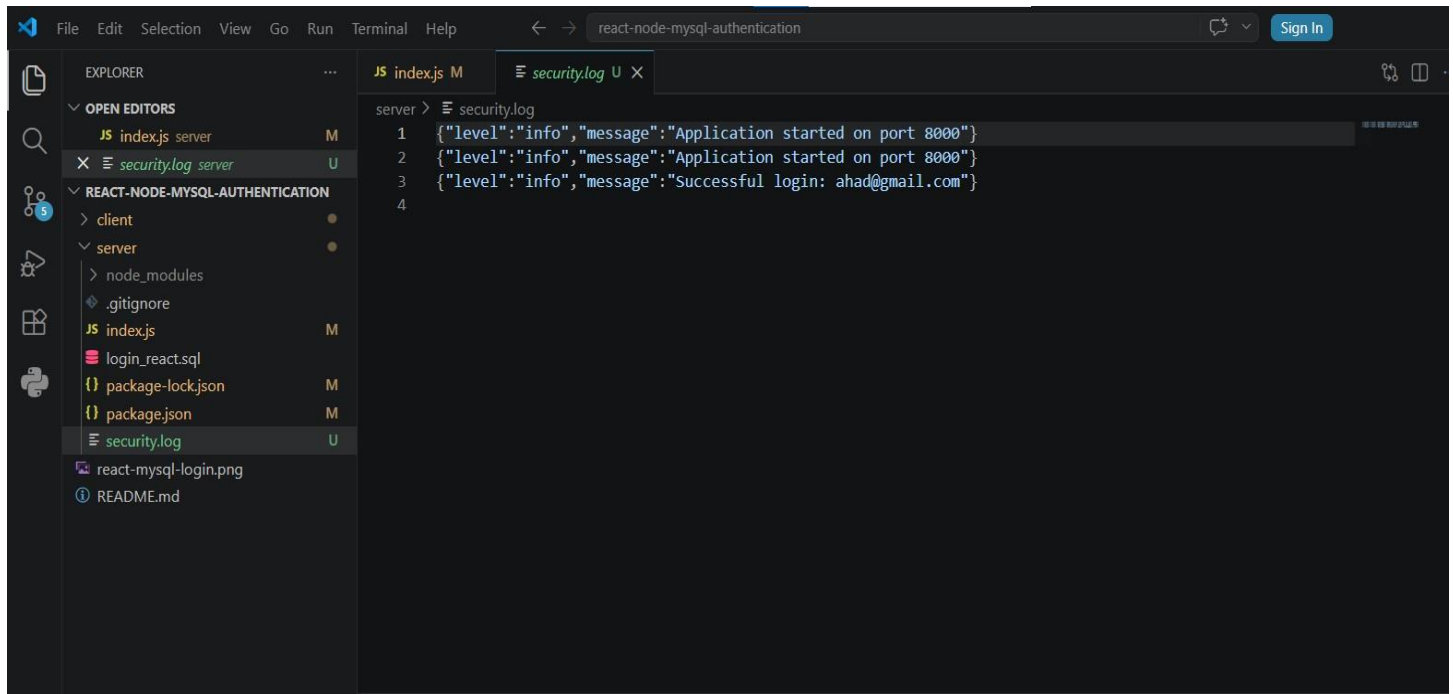
The **Winston** logging library was implemented to provide basic application logging.

Implementation

- Installed the Winston package.
- Configured logging to display messages in the console.
- Created a **security.log** file to store log entries.
- Verified that application startup events were successfully recorded.

📸 Screenshot 4

Winston Logger and security.log



3. Security Best Practices Checklist

Best Practice	Status
Validate all user inputs	✓
Hash and salt passwords using bcrypt	✓
Implement JWT-based authentication	✓
Secure HTTP headers using Helmet	✓
Maintain application logs using Winston	✓